Zetao Zhou

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Education

University College London — London, United Kingdom
Bachelor of Science in Psychology, (Upper Second Class Hons, 2:1) — *Sep 2022 – Jun 2025*

Experience

Software Engineering Intern — Jun 2026 – Sep 2026
Fulcrum GT — *London, United Kingdom*

- Sole developer on a 2-people team building a legal-tech staffing-analytics MVP end-to-end; owned all engineering: architecture, backend, data, AI, and prototype.
- Architected the system and authored a 2-level C4 model (context + container) spanning a Next.js/TypeScript + Tailwind frontend, Python/FastAPI backend, PostgreSQL store, Python ETL sync service, and self-hosted Keycloak SSO (OAuth2).
- Designed a permission-scoped RAG pipeline (vector retrieval over PostgreSQL + pg_embeddings feeding a self-hosted Gemma LLM) to generate cited, audit-logged explanations while keeping privileged client data on-cluster.
- Formulated a transparent matched-cohort statistical model to quantify the margin impact of staffing choices, defining explicit capability, utilization, predicted-margin, and weighted-scoring formulas with per-firm tunable parameters.

Human-AI Interaction Research Assistant — May 2024 – Sep 2024
University College London, Computer Science Department — *London, United Kingdom*

- Redesigned the study graphical interface using Python (Tkinter) to an established codebase via Git.
- Translated complex research and behavioural requirements into structured interface components.
- Integrated LLM APIs into the interface to enable interactive user experiences.
- Built embedded knowledge modules for the GPT-3 model to support task-specific behaviour.
- Conducted an in-person human subject study to collect quantitative and qualitative data from 40 participants within 2 weeks.
- Delivered end-to-end software development lifecycle: requirements interpretation, system design, implementation, testing, deployment, and data analysis.
- Co-authored a paper currently under review: arxiv.org/abs/2411.06950

Projects

VR Multisensory Hardware Development | *Arduino,C♯, Unity* — Sep 2024 – Mar 2025

- Designed and built a hardware-software integrated system using Arduino and Unity, implementing deterministic control logic and safe shutdown mechanisms.
- Developed structured, maintainable C-style code for PWM signal control and automated randomized event sequencing and VR-triggered event logic.
- Engineered trial logic and validation checks to ensure consistent system behaviour.
- Iteratively improved system reliability through pilot testing and controlled evaluation.

Experiment Software Development | *Python, QtDesigner* — Nov 2024 – Dec 2024

- Designed and implemented a 400+ line Python application with PyQt5 interface to automate randomization, data validation, and structured logging.
- Applied testing principles to ensure accurate data capture and reproducible outputs.
- Built modular components to enable extension and long-term maintainability.
- Reduced manual validation effort by automating key submission and checking processes.

Technical Skills

Languages: Java, Python, SQL (PostgresSQL), TypeScript, JavaScript, HTML/CSS, C♯, R
Frameworks: React, Next.js, Node.js, Tailwind CSS, FastAPI
Developer Tools: Git, Docker, Keycloak, GCP, VS Code, Visual Studio, JetBrains (PyCharm/IntelliJ)
Libraries: pandas, NumPy, Matplotlib, PyQt5, TKinter
Concepts: REST APIs, LLM integration / RAG, system design, C4